HEALTHCARE ANALYTICS PORTFOLIO

Healthcare Waitlist Analysis

Uncovering bottlenecks in hospital patient flow through data visualization and intelligent dashboard design.

ANALYST
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PRIMARY TOOL
Power BI

DATA PREP
Excel / Power Query

Problem Statement

The Issue

Hospital management struggled to track patient wait times across departments. Without a unified view, delays went unnoticed, resource allocation was reactive, and patients experienced increasingly long wait times with no clear accountability.

- No Centralized Tracking
- Reactive Decisions
- Patient Dissatisfaction

The Goal

Create an interactive visual tool to identify delays and improve patient flow. The dashboard would enable management to spot bottlenecks by specialty, age profile, and time band — turning raw waitlist data into actionable insight.

- Visual Dashboard
- Identify Delays
- Improve Flow

629K	563K	+11.7%
Latest Month Waitlist	Previous Year Same Month	Year-over-Year Increase

Data Cleaning

STEP 1

Loading Data

Imported and combined **Inpatient** and **Outpatient** Excel files into a unified data model within Power BI.

2 Excel sources merged

STEP 2

Fixing Errors

Used **Power Query** to resolve *"Model object name"* conflicts and *"Date Locale"* mismatches between the two source files.

2 critical errors resolved

STEP 3

Formatting

Standardized **Age_Profile** as text data type and converted **Archive_Date** to proper date format for accurate time-series analysis.

2 columns standardized

Inpatient.xlsx | Outpatient.xlsx → Power Query → Clean Unified Model

Key Insights

Average | Median

629K
Latest Month Wait List
↑ 11.7% YoY

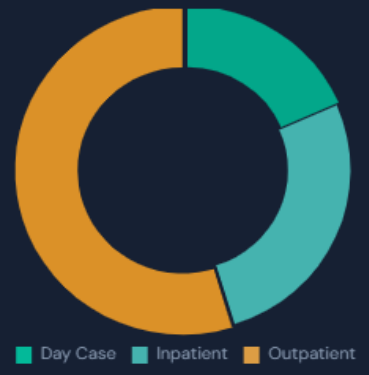
563K
PY Latest Month Wait List
— Baseline

6.8
Avg Wait (Months)
↑ vs 5.2 PY

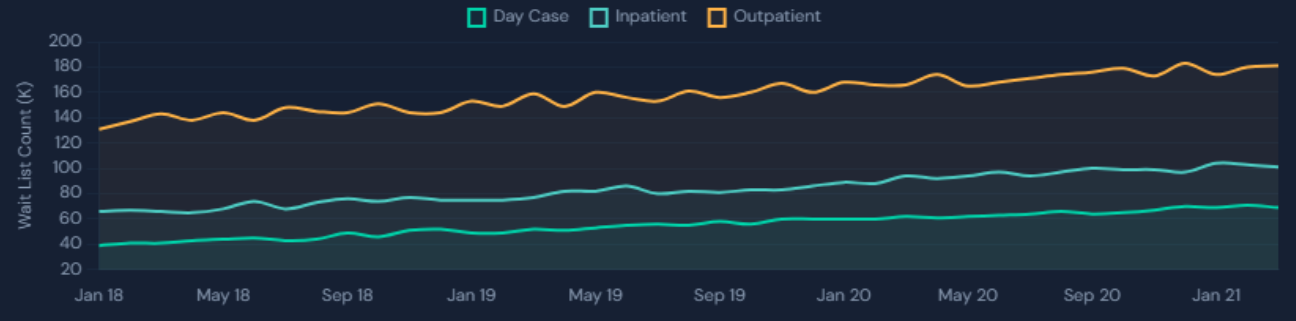
3
Case Types Tracked
Day / In / Out

Avg Wait List by Time_Bands and Age_Profile

Avg Wait List by Case_Type



Sum of Total by Archive_Date and Case_Type



Top 5 Wait List by Specialty

1	Accident & Emergency	69.5
2	Orthopaedics	58.3
3	Gynaecology	42.7
4	Ophthalmology	38.1
5	ENT	31.4

Granular Analysis

All | Orthopaedics | Gynaecology | Ophthalmology | ENT | Gen. Surgery

629K
Latest Month Wait List

563K
PY Latest Month

10
Specialties Analyzed

38
Months Covered

Sum of Total by Specialty_Name

Wait List by Archive_Date and Case_Type

ARCHIVE DATE	DAY CASE	INPATIENT	OUTPATIENT	TOTAL
Mar 2018	32	57	124	213
Jun 2018	36	59	135	230
Sep 2018	40	55	131	226
Dec 2018	36	59	150	245
Mar 2019	41	72	135	248
Jun 2019	41	66	144	251
Sep 2019	46	69	159	274
Dec 2019	45	75	170	290
Mar 2020	38	65	157	260
Jun 2020	41	56	122	219
Sep 2020	49	71	169	289
Dec 2020	51	85	176	312
Mar 2021	54	79	202	335
Total	**550**	**868**	**1,974**	**3,392**